

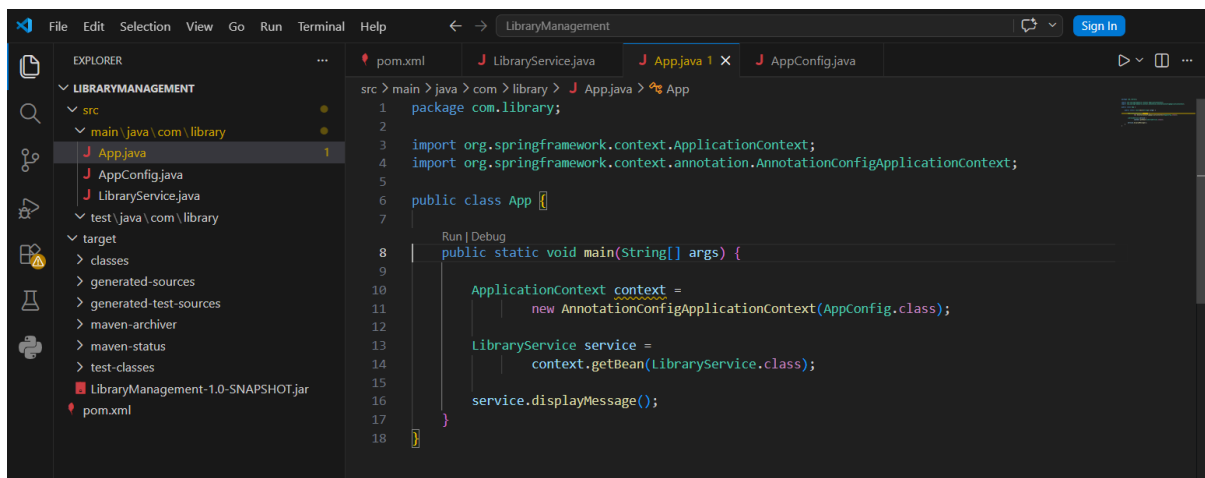
COGNIZANT

Mandatory Hands-on Exercises

TOPIC: Spring Core and Maven

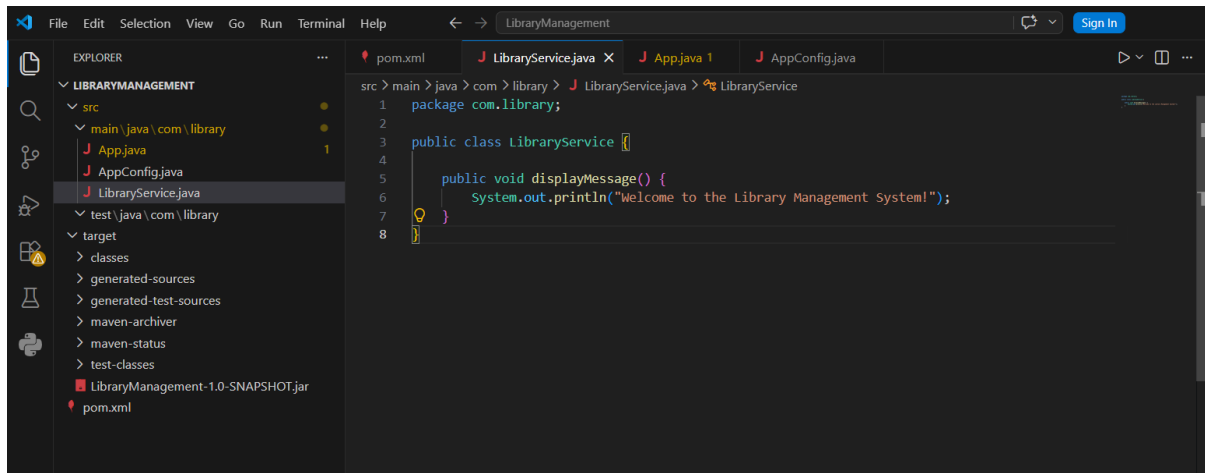
Exercise 1: Configuring a Basic Spring Application

CODE



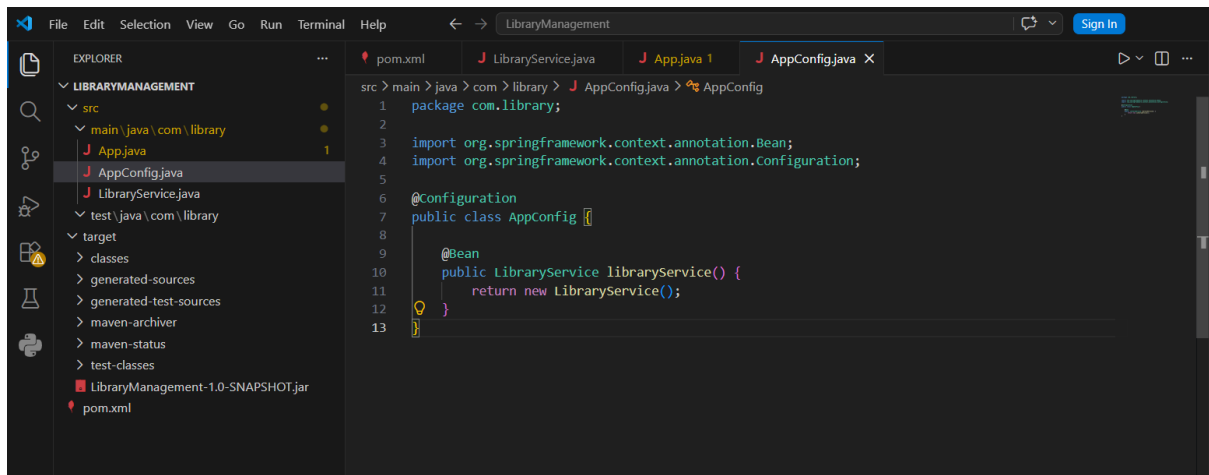
The screenshot shows an IDE window for a project named "LibraryManagement". The Explorer panel on the left shows the project structure: src > main > java > com > library. The main editor displays the code for App.java:

```
1 package com.library;
2
3 import org.springframework.context.ApplicationContext;
4 import org.springframework.context.annotation.AnnotationConfigApplicationContext;
5
6 public class App {
7
8     Run | Debug
9     public static void main(String[] args) {
10
11         ApplicationContext context =
12             new AnnotationConfigApplicationContext(AppConfig.class);
13
14         LibraryService service =
15             context.getBean(LibraryService.class);
16
17         service.displayMessage();
18     }
19 }
```

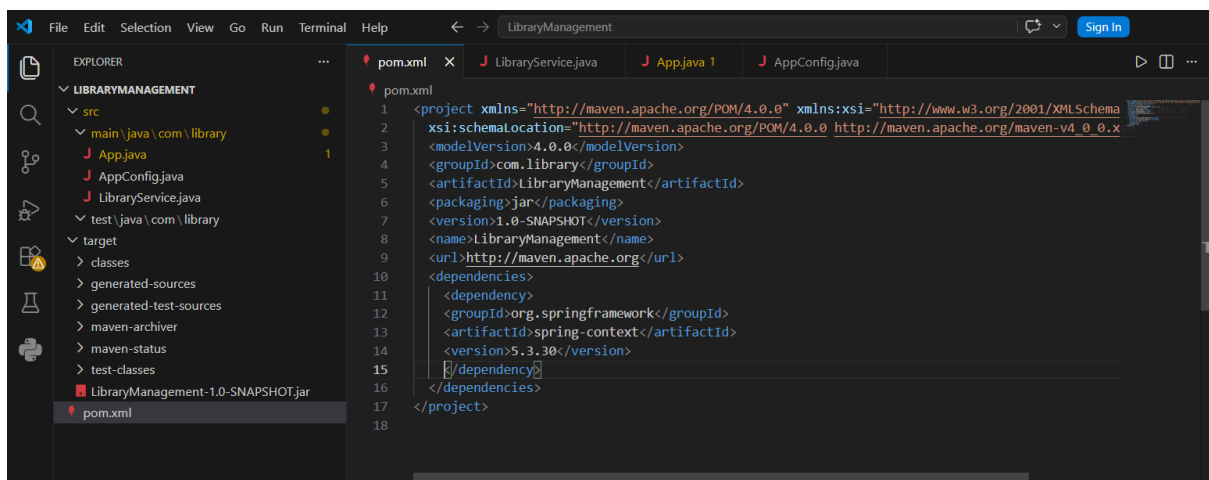


The screenshot shows the same IDE window, but now the LibraryService.java file is selected in the Explorer panel and displayed in the main editor:

```
1 package com.library;
2
3 public class LibraryService {
4
5     public void displayMessage() {
6         System.out.println("Welcome to the Library Management System!");
7     }
8 }
```



```
src > main > java > com > library > AppConfig.java > AppConfig
1 package com.library;
2
3 import org.springframework.context.annotation.Bean;
4 import org.springframework.context.annotation.Configuration;
5
6 @Configuration
7 public class AppConfig {
8
9     @Bean
10    public LibraryService libraryService() {
11        return new LibraryService();
12    }
13 }
```



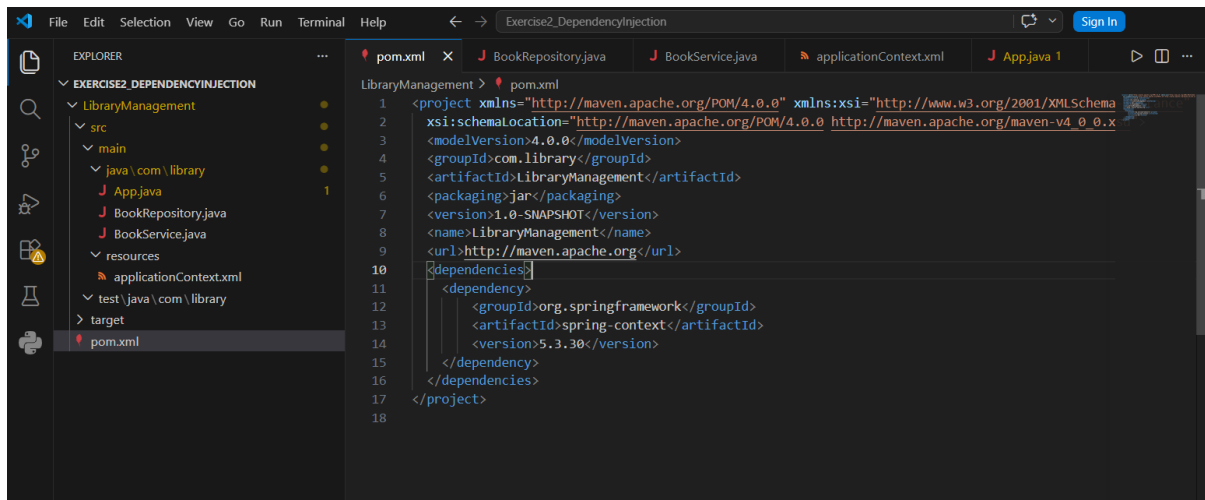
```
1 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema"
2 xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.x
3 <modelVersion>4.0.0</modelVersion>
4 <groupId>com.library</groupId>
5 <artifactId>LibraryManagement</artifactId>
6 <packaging>jar</packaging>
7 <version>1.0-SNAPSHOT</version>
8 <name>LibraryManagement</name>
9 <url>http://maven.apache.org</url>
10 <dependencies>
11 <dependency>
12 <groupId>org.springframework</groupId>
13 <artifactId>spring-context</artifactId>
14 <version>5.3.30</version>
15 </dependency>
16 </dependencies>
17 </project>
18
```

OUTPUT

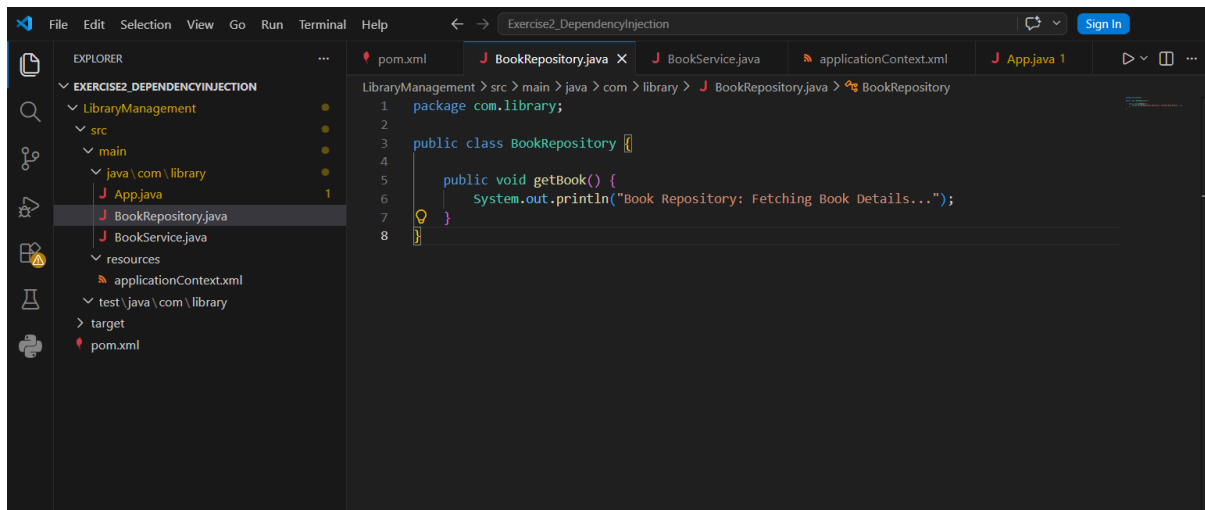
```
C:\Users\Khushi Ahi\OneDrive\Desktop\Exercise1_BasicSpringApp\LibraryManagement>mvn compile exec:java
-Dexec.mainClass="com.library.App"
[INFO] Scanning for projects...
[INFO] -----< com.library:LibraryManagement >-----
[INFO] Building LibraryManagement 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- resources:3.4.0:resources (default-resources) @ LibraryManagement ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform
dependent!
[INFO] skip non existing resourceDirectory C:\Users\Khushi Ahi\OneDrive\Desktop\Exercise1_BasicSpring
App\LibraryManagement\src\main\resources
[INFO] --- compiler:3.15.0:compile (default-compile) @ LibraryManagement ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] --- exec:3.6.3:java (default-cli) @ LibraryManagement ---
Welcome to the Library Management System!
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 1.599 s
[INFO] Finished at: 2026-07-04T14:38:16+05:30
[INFO] -----
C:\Users\Khushi Ahi\OneDrive\Desktop\Exercise1_BasicSpringApp\LibraryManagement>
```

Exercise 2: Implementing Dependency Injection

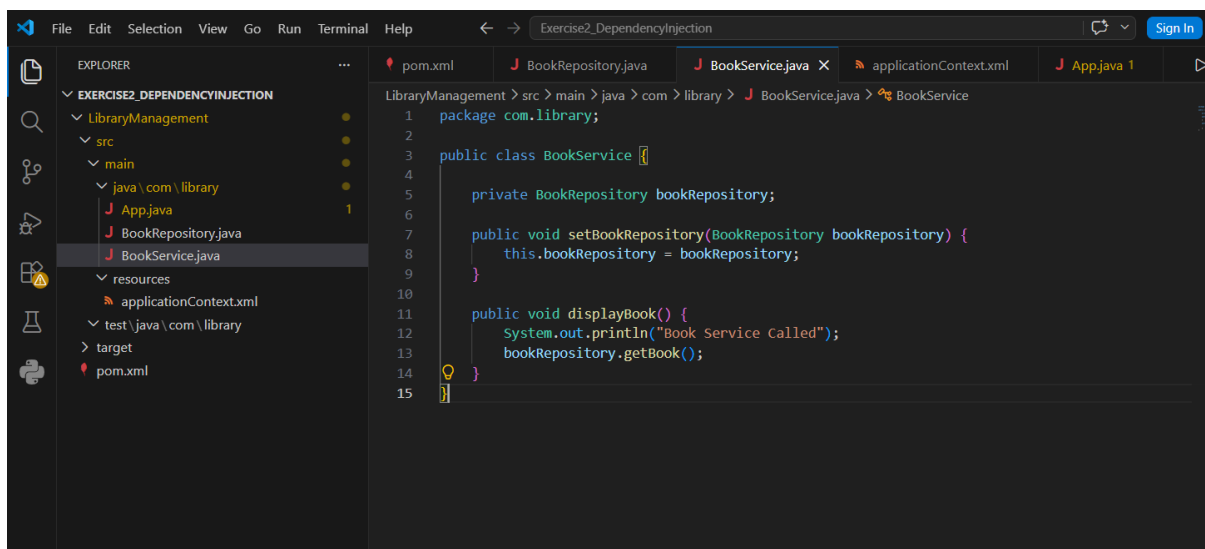
CODE



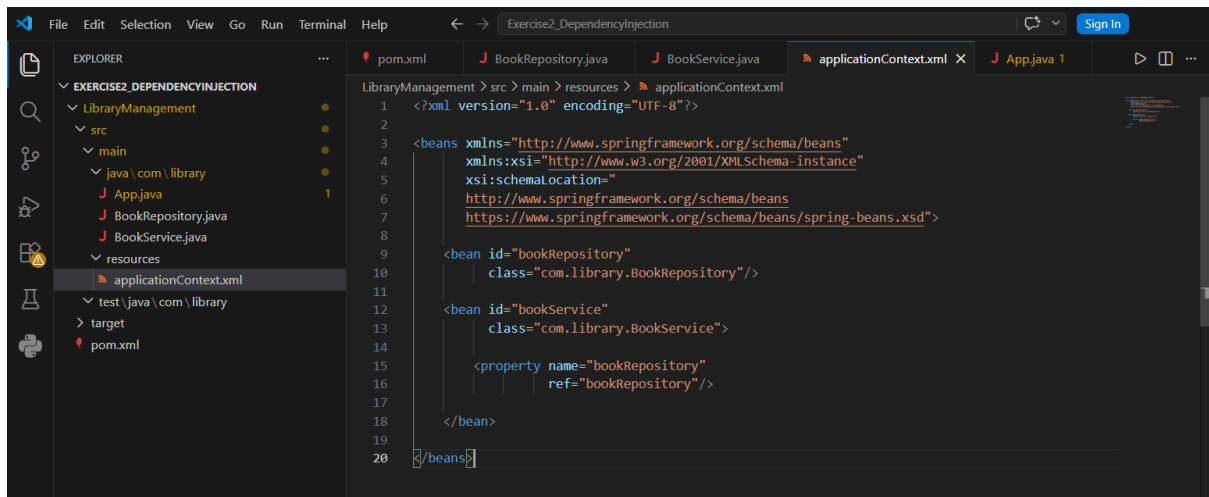
```
1 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema"
2 xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.x
3 <modelVersion>4.0.0</modelVersion>
4 <groupId>com.library</groupId>
5 <artifactId>LibraryManagement</artifactId>
6 <packaging>jar</packaging>
7 <version>1.0-SNAPSHOT</version>
8 <name>LibraryManagement</name>
9 <url>http://maven.apache.org</url>
10 <dependencies>
11 <dependency>
12 <groupId>org.springframework</groupId>
13 <artifactId>spring-context</artifactId>
14 <version>5.3.30</version>
15 </dependency>
16 </dependencies>
17 </project>
18
```



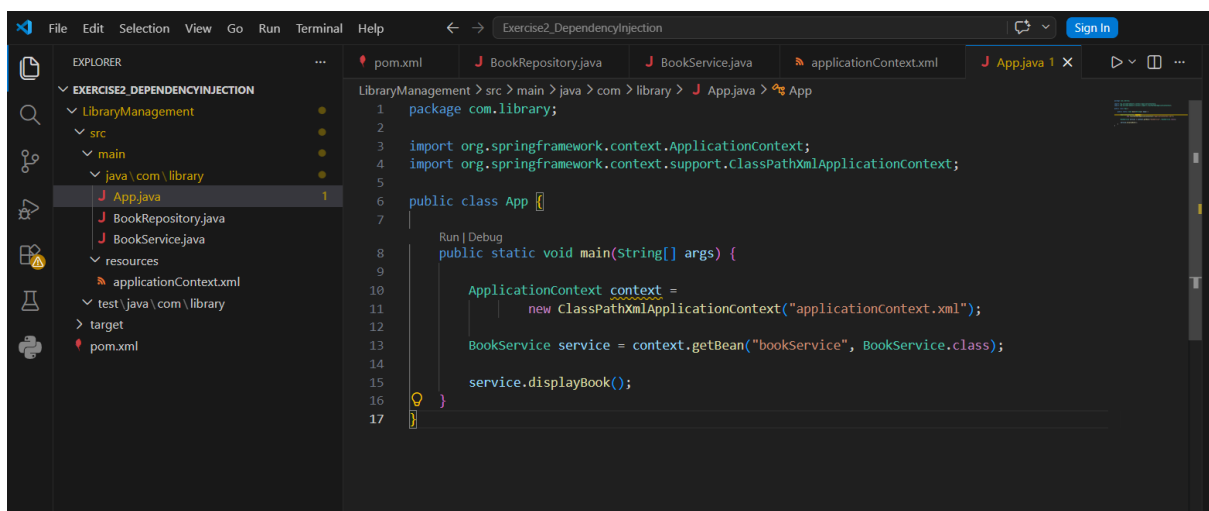
```
1 package com.library;
2
3
4 public class BookRepository {
5
6     public void getBook() {
7         System.out.println("Book Repository: Fetching Book Details...");
8     }
9 }
```



```
1 package com.library;
2
3 public class BookService {
4
5     private BookRepository bookRepository;
6
7     public void setBookRepository(BookRepository bookRepository) {
8         this.bookRepository = bookRepository;
9     }
10
11     public void displayBook() {
12         System.out.println("Book Service Called");
13         bookRepository.getBook();
14     }
15 }
```

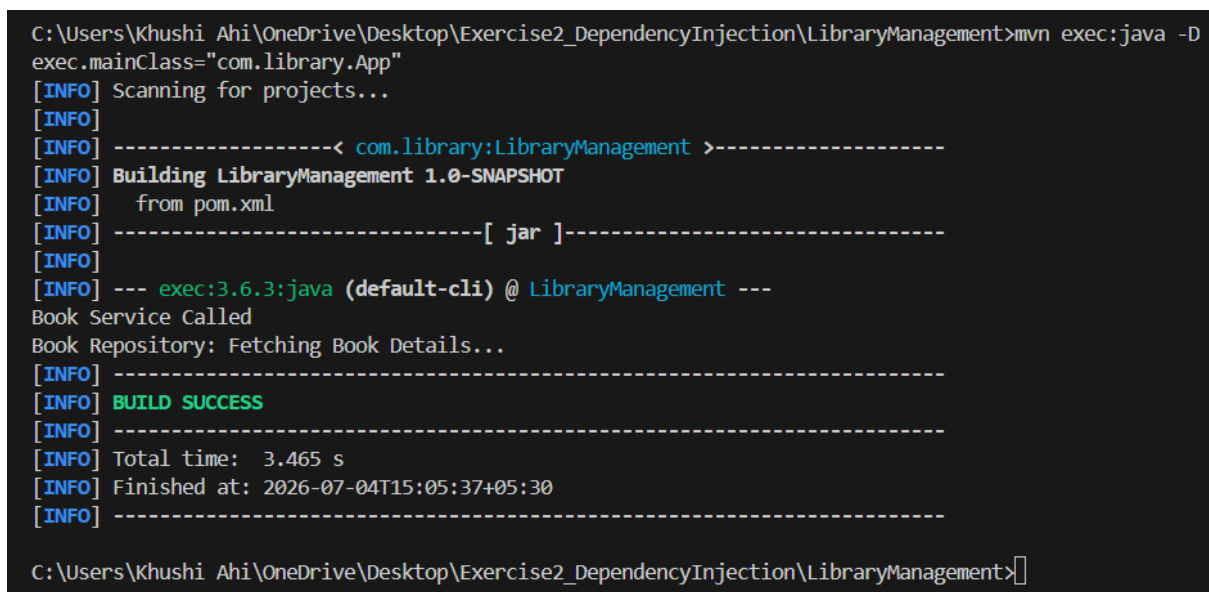


```
1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <beans xmlns="http://www.springframework.org/schema/beans"
4       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5       xsi:schemaLocation="
6         http://www.springframework.org/schema/beans
7         https://www.springframework.org/schema/beans/spring-beans.xsd">
8
9   <bean id="bookRepository"
10        class="com.library.BookRepository"/>
11
12   <bean id="bookService"
13        class="com.library.BookService">
14
15     <property name="bookRepository"
16              ref="bookRepository"/>
17
18   </bean>
19
20 </beans>
```



```
1 package com.library;
2
3 import org.springframework.context.ApplicationContext;
4 import org.springframework.context.support.ClassPathXmlApplicationContext;
5
6 public class App {
7
8     Run | Debug
9     public static void main(String[] args) {
10
11         ApplicationContext context =
12             new ClassPathXmlApplicationContext("applicationContext.xml");
13
14         BookService service = context.getBean("bookService", BookService.class);
15
16         service.displayBook();
17     }
18 }
```

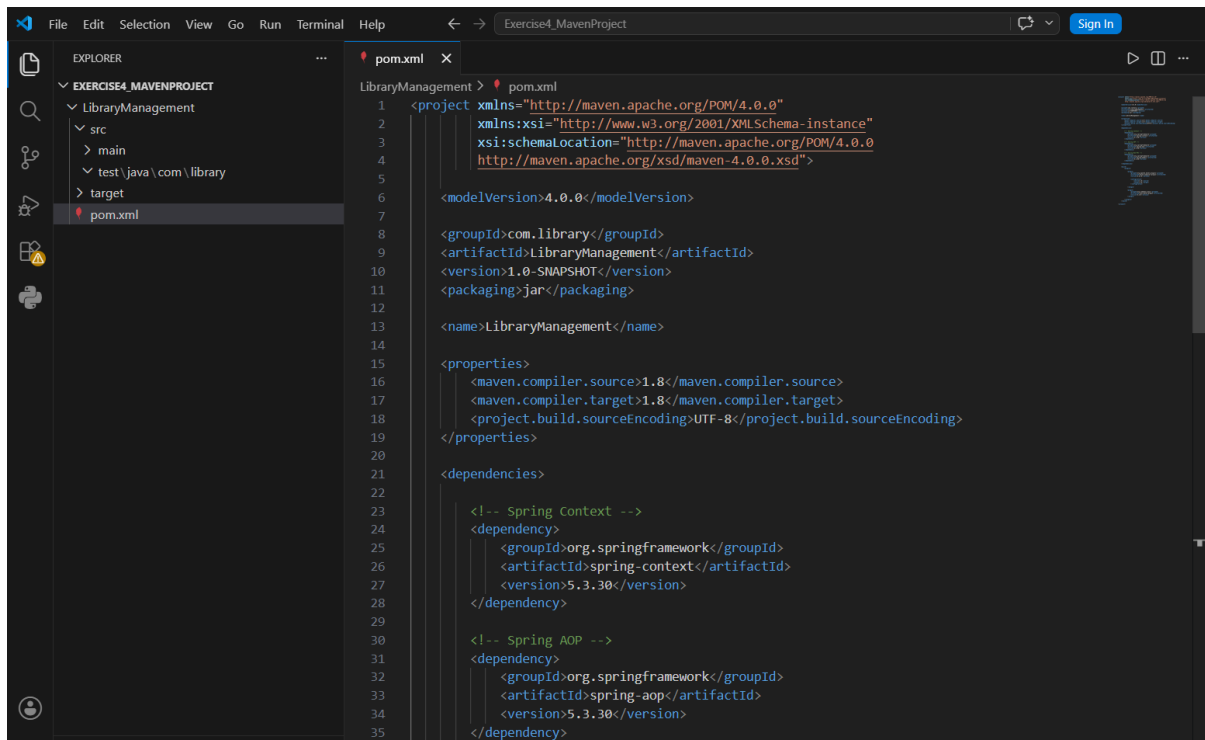
OUTPUT



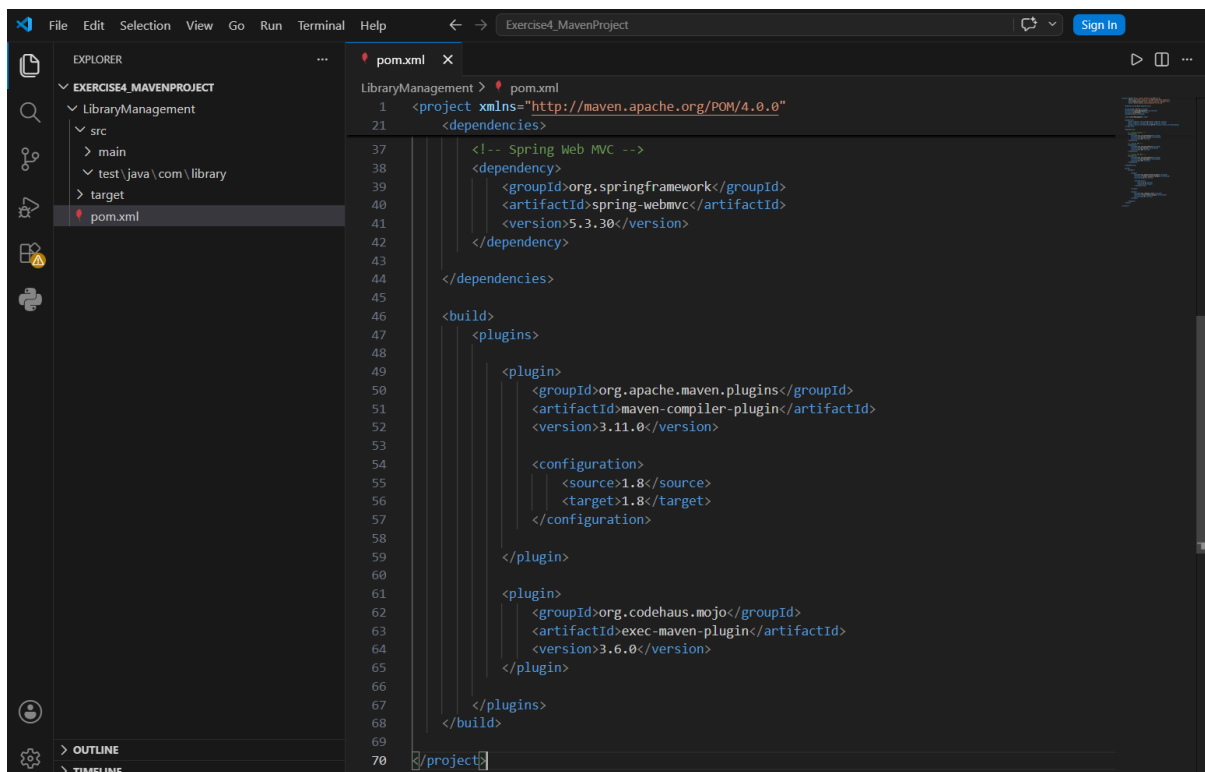
```
C:\Users\Khushi Ahi\OneDrive\Desktop\Exercise2_DependencyInjection\LibraryManagement>mvn exec:java -D
exec.mainClass="com.library.App"
[INFO] Scanning for projects...
[INFO] -----< com.library:LibraryManagement >-----
[INFO] Building LibraryManagement 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- exec:3.6.3:java (default-cli) @ LibraryManagement ---
Book Service Called
Book Repository: Fetching Book Details...
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 3.465 s
[INFO] Finished at: 2026-07-04T15:05:37+05:30
[INFO] -----
C:\Users\Khushi Ahi\OneDrive\Desktop\Exercise2_DependencyInjection\LibraryManagement>
```

Exercise 4: Creating and Configuring a Maven Project

CODE

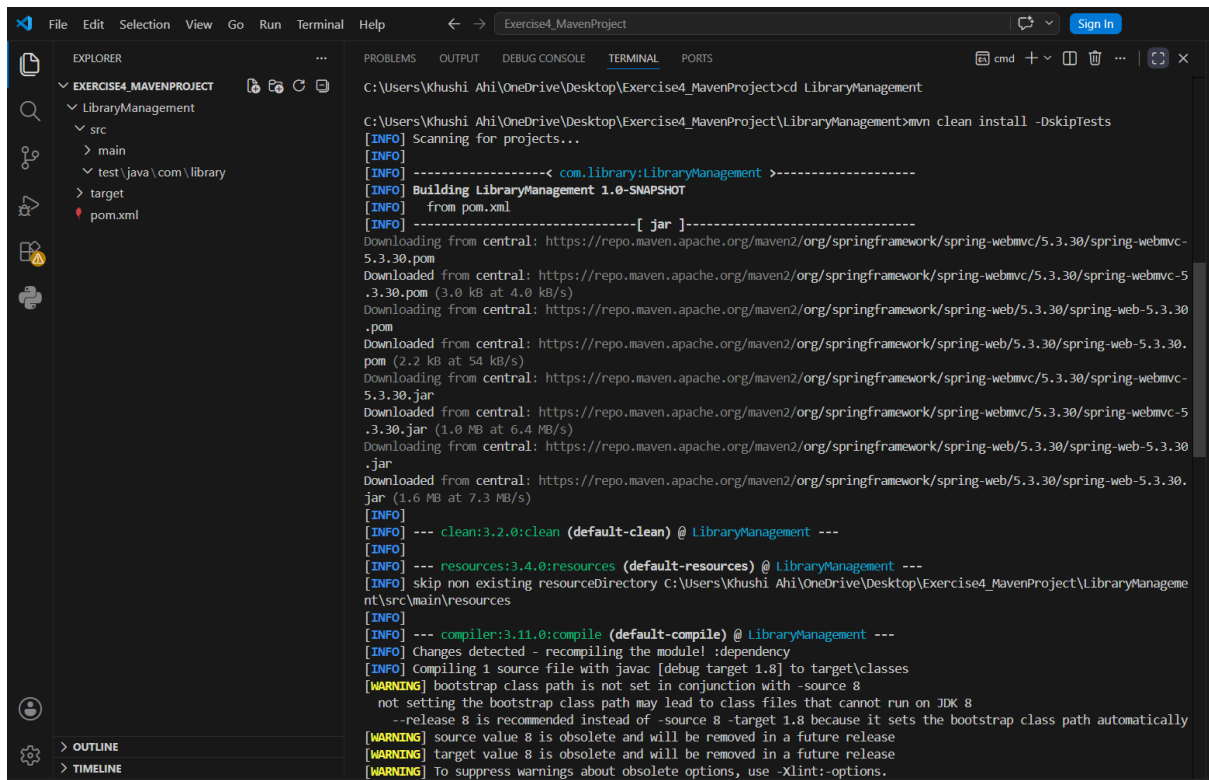


```
1 <project xmlns="http://maven.apache.org/POM/4.0.0"
2         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
4         http://maven.apache.org/xsd/maven-4.0.0.xsd">
5
6     <modelVersion>4.0.0</modelVersion>
7
8     <groupId>com.library</groupId>
9     <artifactId>LibraryManagement</artifactId>
10    <version>1.0-SNAPSHOT</version>
11    <packaging>jar</packaging>
12
13    <name>LibraryManagement</name>
14
15    <properties>
16        <maven.compiler.source>1.8</maven.compiler.source>
17        <maven.compiler.target>1.8</maven.compiler.target>
18        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
19    </properties>
20
21    <dependencies>
22
23        <!-- Spring Context -->
24        <dependency>
25            <groupId>org.springframework</groupId>
26            <artifactId>spring-context</artifactId>
27            <version>5.3.30</version>
28        </dependency>
29
30        <!-- Spring AOP -->
31        <dependency>
32            <groupId>org.springframework</groupId>
33            <artifactId>spring-aop</artifactId>
34            <version>5.3.30</version>
35        </dependency>
36    </dependencies>
```



```
1 <project xmlns="http://maven.apache.org/POM/4.0.0"
2         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
4         http://maven.apache.org/xsd/maven-4.0.0.xsd">
5
6     <modelVersion>4.0.0</modelVersion>
7
8     <groupId>com.library</groupId>
9     <artifactId>LibraryManagement</artifactId>
10    <version>1.0-SNAPSHOT</version>
11    <packaging>jar</packaging>
12
13    <name>LibraryManagement</name>
14
15    <properties>
16        <maven.compiler.source>1.8</maven.compiler.source>
17        <maven.compiler.target>1.8</maven.compiler.target>
18        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
19    </properties>
20
21    <dependencies>
22
23        <!-- Spring Web MVC -->
24        <dependency>
25            <groupId>org.springframework</groupId>
26            <artifactId>spring-webmvc</artifactId>
27            <version>5.3.30</version>
28        </dependency>
29    </dependencies>
30
31    <build>
32        <plugins>
33
34            <plugin>
35                <groupId>org.apache.maven.plugins</groupId>
36                <artifactId>maven-compiler-plugin</artifactId>
37                <version>3.11.0</version>
38
39                <configuration>
40                    <source>1.8</source>
41                    <target>1.8</target>
42                </configuration>
43            </plugin>
44
45            <plugin>
46                <groupId>org.codehaus.mojo</groupId>
47                <artifactId>exec-maven-plugin</artifactId>
48                <version>3.6.0</version>
49            </plugin>
50
51        </plugins>
52    </build>
53</project>
```

OUTPUT



The screenshot shows an IDE with a terminal window open. The Explorer pane on the left shows the project structure: EXERCISE4_MAVENPROJECT > LibraryManagement > src > main > test\java\com\library > target > pom.xml. The terminal window shows the command `C:\Users\Khushi Ahi\OneDrive\Desktop\Exercise4_MavenProject>cd LibraryManagement` and the subsequent Maven build process. The output includes downloading dependencies from the central repository, cleaning the project, and compiling the source code. The build is successful, and the output ends with `[INFO] BUILD SUCCESS`.

```
C:\Users\Khushi Ahi\OneDrive\Desktop\Exercise4_MavenProject>cd LibraryManagement
C:\Users\Khushi Ahi\OneDrive\Desktop\Exercise4_MavenProject\LibraryManagement>mvn clean install -DskipTests
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.library:LibraryManagement >-----
[INFO] Building LibraryManagement 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/springframework/spring-webmvc/5.3.30/spring-webmvc-5.3.30.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/spring-webmvc/5.3.30/spring-webmvc-5.3.30.pom (3.0 kB at 4.0 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/springframework/spring-web/5.3.30/spring-web-5.3.30.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/spring-web/5.3.30/spring-web-5.3.30.pom (2.2 kB at 54.0 kB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/springframework/spring-webmvc/5.3.30/spring-webmvc-5.3.30.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/spring-webmvc/5.3.30/spring-webmvc-5.3.30.jar (1.0 MB at 6.4 MB/s)
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/org/springframework/spring-web/5.3.30/spring-web-5.3.30.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/springframework/spring-web/5.3.30/spring-web-5.3.30.jar (1.6 MB at 7.3 MB/s)
[INFO] --- clean:3.2.0:clean (default-clean) @ LibraryManagement ---
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ LibraryManagement ---
[INFO] skip non existing resourceDirectory C:\Users\Khushi Ahi\OneDrive\Desktop\Exercise4_MavenProject\LibraryManagement\src\main\resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ LibraryManagement ---
[INFO] Changes detected - recompiling the module! :dependency
[INFO] compiling 1 source file with javac [debug target 1.8] to target\classes
[WARNING] bootstrap class path is not set in conjunction with -source 8
not setting the bootstrap class path may lead to class files that cannot run on JDK 8
--release 8 is recommended instead of -source 8 -target 1.8 because it sets the bootstrap class path automatically
[WARNING] source value 8 is obsolete and will be removed in a future release
[WARNING] target value 8 is obsolete and will be removed in a future release
[WARNING] To suppress warnings about obsolete options, use -Xlint:-options.
```

```
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ LibraryManagement ---
[INFO] skip non existing resourceDirectory C:\Users\Khushi Ahi\OneDrive\Desktop\Exercise4_MavenProject\LibraryManagement\src\test\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ LibraryManagement ---
[INFO] Changes detected - recompiling the module! :dependency
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ LibraryManagement ---
[INFO] Tests are skipped.
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ LibraryManagement ---
[INFO] Building jar: C:\Users\Khushi Ahi\OneDrive\Desktop\Exercise4_MavenProject\LibraryManagement\target\LibraryManagement-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- install:3.1.4:install (default-install) @ LibraryManagement ---
[INFO] Installing C:\Users\Khushi Ahi\OneDrive\Desktop\Exercise4_MavenProject\LibraryManagement\pom.xml to C:\Users\Khushi Ahi\.m2\repository\com\library\LibraryManagement\1.0-SNAPSHOT\LibraryManagement-1.0-SNAPSHOT.pom
[INFO] Installing C:\Users\Khushi Ahi\OneDrive\Desktop\Exercise4_MavenProject\LibraryManagement\target\LibraryManagement-1.0-SNAPSHOT.jar to C:\Users\Khushi Ahi\.m2\repository\com\library\LibraryManagement\1.0-SNAPSHOT\LibraryManagement-1.0-SNAPSHOT.jar
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 4.231 s
[INFO] Finished at: 2026-07-04T15:17:11+05:30
[INFO] -----

C:\Users\Khushi Ahi\OneDrive\Desktop\Exercise4_MavenProject\LibraryManagement>
```